19 The diagram shows triangle ABC

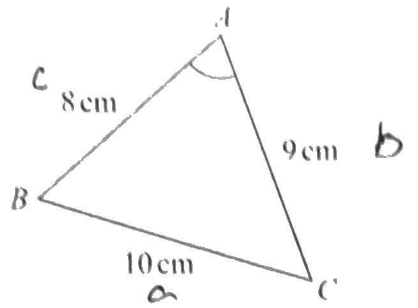

Diagram **NOT** accurately drawn

$AB = 8\text{ cm}$ $\quad BC = 10\text{ cm}$ $\quad CA = 9\text{ cm}$

Work out the size of angle BAC
Give your answer correct to one decimal place.

Cosrule

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$\cos A = \frac{9^2+8^2-10^2}{2\times 8\times 9}$$

$$A = \cos^{-1}\left(\frac{5}{16}\right)$$

$$A = 71.790$$

$$\underline{A = 71.8^\circ}$$

71.8 °

(Total for Question 19 is 3 marks)